Holistic Response Lifting for Weakly Supervised Land-Cover Classification

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***Abstract*—This article proposes a holistic response lifting (HoLift) scheme to tackle the issue of region ambiguity in class activation maps (CAMs) for weakly supervised remote sensing land-cover classification (WSLC). In general, the ambiguously responding zones in CAMs are annotated as the background in the pseudolabel for object-centric natural images. However, it is infeasible to postprocess the region ambiguity in CAMs at the stage of computing pseudolabels for WSLC. As different land covers are seamlessly connected without background definition in satellite images, treating ambiguous responses as the background will destroy the stuff-level continuity in land-cover pseudolabels. Therefore, the ambiguous zones must be directly eliminated from land-cover CAMs rather than classified as the background in pseudolabels. To this end, our HoLift scheme is designed for the following: 1) force seamless area responses in CAMs with a holistic continuity lifting loss and 2) reduce response ambiguity in border areas via a cross-image global consistency constraint. Specifically, the pixel-level holistic continuity lifting loss is applied to multiclass CAMs to promote high total responses, reducing the ambiguous areas that lead to discontinuous pseudolabels. Meanwhile, the cross-image global consistency constraint facilitates the learning of comprehensive land-cover features by optimizing the distribution of regional features across images, thereby improving classification reliability near land-cover boundaries. With HoLift, we obtain CAMs free of large-area ambiguous transition zones. Extensive experiments reveal that our HoLift scheme significantly improves recall of CAMs by at least 16.08% and achieves state-of-the-art performance on two challenging datasets. Our code is available at https://github.com/Geo-Tell/HoLift**

***Index Terms*—Class activation map (CAM), holistic response lifting (HoLift), land-cover classification, remote sensing, weakly supervised learning.**

I. Introduction

LAND-COVER classification aims to describe the Earth's surface and is fundamental to extensive applications, including urban planning [1], [2], environmental monitoring [3], and global-scale geographical analysis [4], [5]. As remote sensing land covers show highly heterogeneous appearances around the world, training land-cover classification models requires numerous samples with pixel-level annotations and diverse geographic distribution to guarantee global generalization of models [6]. However, acquiring pixelwise labels for high-resolution remote sensing images is time-consuming and labor-intensive [7]. To avoid the reliance on expensive pixel-level annotations, increasing attention has been recently paid to weakly supervised land-cover classification (WSLC) using labor-friendly image-level labels.

In recent years, WSLC has been studied with the general weakly supervised semantic segmentation (WSSS) pipeline designed for natural data [8], [9]. A typical WSSS pipeline begins by training an image-level classification network, which generates class activation maps (CAMs). These CAMs are then used to produce pixel-level pseudolabels, which are subsequently employed to train semantic segmentation models. However, we find that current efforts often encounter the issue of ambiguously responding, which destroys the land-cover continuity in pseudolabels derived from CAMs [10] for WSLC. Actually, due to the lack of precise localization guidance in weak supervision, the classification network in WSSS tends to focus on salient local patterns. As a result, the generated CAMs only highlight sparse object regions, leaving considerable low-response areas, as highlighted in Fig. 1(a). These regions are ambiguous in determining the category, and the response ambiguity in objects leads to incomplete pseudolabels for the training of semantic segmentation models.

Various solutions have been proposed to reduce the impact of ambiguous zones on pseudolabels, such as the erasing-based refinding [11], [12], [13], [14], CAM ensemble [15], [16], [17], and affinity-based random walk [18], [19], [20]. These WSSS approaches generally aggregate multiple CAMs that pay attention to different local salient patterns, which forces the region ambiguity to mainly occur in the nonsalient or background areas. In this case, it is feasible to obtain object-focused pseudolabels by filtering ambiguously responding regions in the fused CAM with a fixed threshold. Albeit effective for natural images, the general WSSS pipeline fails in tackling the ambiguously responding regions in land-cover CAMs. **This is because the target of WSLC in remote sensing images fundamentally differs from that of the general pipeline of WSSS in natural images.**

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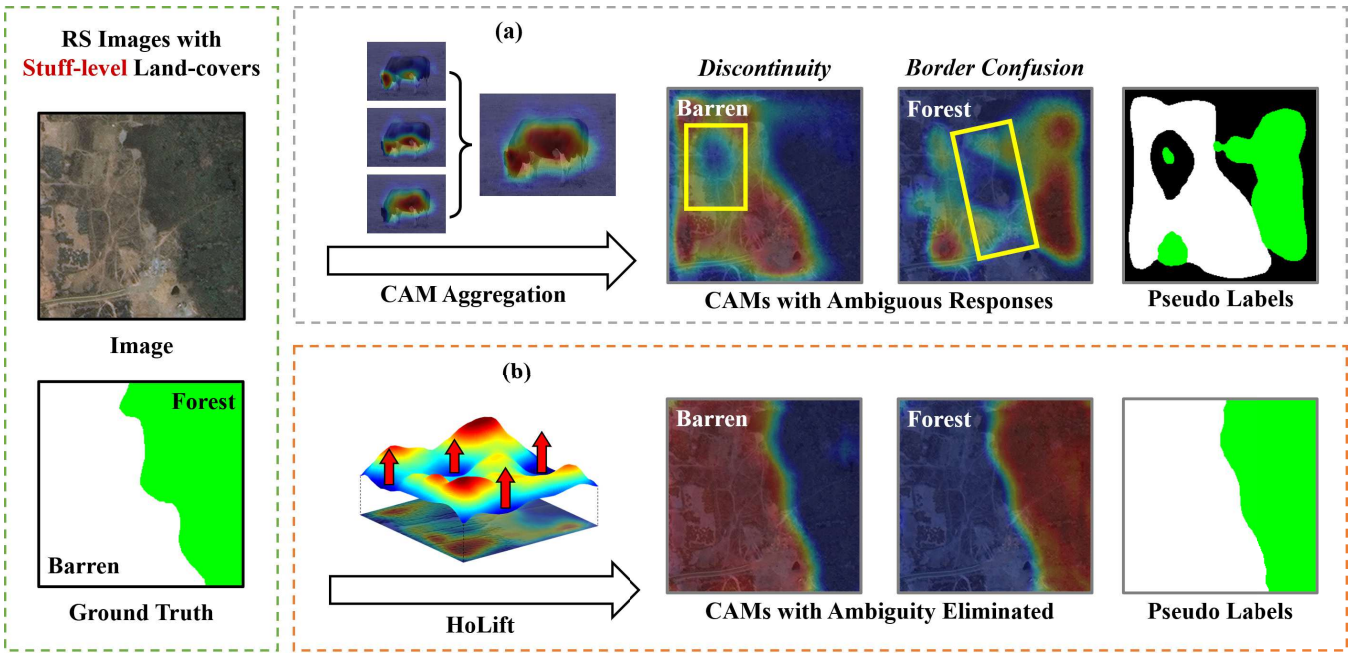


Fig. 1. Comparison between the CAM aggregation and our HoLift scheme in weakly supervised land-cover classification. (a) Directly applying CAM aggregation devised for objects in natural images to land covers in remote sensing images. (b) Applying our HoLift scheme devised for stuff-level land covers.

Specifically, the general WSSS target in natural data is *object-level*, while the land cover in satellite images is *stuff-level*. Objects in natural images are countable individuals with regular shapes and local salient patterns. Differently, stuff is a term used in panoptic segmentation [21] to describe amorphous and uncountable regions with nonsalient materials, such as grass and water. For regular objects with constant components, fusing locally responding CAMs can roughly combine different object parts and shrink ambiguous zones to nonsalient or background areas. However, as stuff-level land covers lack representative components and well-defined shapes, aggregating CAMs that focus on salient local patterns will encounter difficulties in holistically describing land cover, thus leaving ambiguous responses inside land-cover interiors and around border areas. Considering that different land covers are seamlessly joined without foreground–background distinction [8], [22], we find that simply annotating ambiguously responding zones as the background will severely break the holistic land-cover continuity in pseudolabels.

As illustrated in Fig. 1(a), the CAM aggregation strategy successfully covers most of the cow, but incurs massive transition zones to land-cover interiors. In addition, due to the absence of pixel-level supervision, aggregated CAMs have difficulty in accurately capturing object details around object boundaries. As the land covers are seamlessly connected in remote sensing images, areas around joint boundaries suffer from response ambiguity, exhibiting low responses across all classes or high responses shared by multiple classes. In Fig. 1(a), even parts of the barren are assigned with high responses in the forest region due to the similarity between local patterns of different classes. These problems inevitably incur annotation discontinuity and semantic errors in the pseudolabel.

Based on the above observation, we believe that the key to high-quality land-cover pseudolabels is to reduce the ambiguous areas in the CAMs with awareness of the global stuff representation instead of salient local patterns. The ambiguous areas are intrinsically caused by the image-level supervision that introduces high tolerance for uncertain pixel-level responses. Therefore, we propose a holistic response lifting scheme (HoLift) to teach a deep image-level classifier to assign each pixel with a high response in its CAMs. Our HoLift works via two losses: 1) a holistic continuity lifting loss and 2) a cross-image global consistency loss. The holistic continuity lifting loss forces every pixel to be noticed with high attention by at least one land-cover category in CAMs, thereby strictly suppressing uncertain responses. Meanwhile, the cross-image global consistency loss builds representation consistency across images for every land-cover category and thus guarantees representation discrimination for clear-cut boundaries between areas of different land-cover categories. By deploying the HoLift strategy with a deep network that simultaneously performs image-level classification and CAM computation, we directly obtain a holistically responding CAM with boundaries for each land cover like in Fig. 1(b), eliminating the need for postprocessing steps like affinity propagation and random walk. Such high-quality CAMs significantly benefit pseudoannotation generation with high area continuity and clear interclass separation.

The contributions are summarized as follows.

1) We introduce a HoLift scheme to generate CAMs that seamlessly describe stuff-level segmentation targets with clear-cut boundaries for weakly supervised land-cover classification in remote sensing images.
2) We build an image-level classifier with HoLift, namely HoLiftNet, to generate high-quality land-cover

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pseudolabels for training a given segmentation model.

3) Extensive experiments are conducted on two challenging land-cover classification benchmarks and the results demonstrate the effectiveness of our HoLift with the recall of CAMs improved by at least 16.08% and the state-of-the-art performance achieved by HoLiftNet.

II. Related Works

A. Weakly Supervised Semantic Segmentation

Semantic segmentation with image-level weak supervision seeks to reduce the burden of pixelwise annotation. Its application in natural scenes has recently attracted much attention. Most methods first generate CAMs [10] from classification networks to locate objects and then predict pixel-level pseudolabels from CAMs for training a semantic segmentation network. However, the generated CAMs usually can only activate sparse discriminative regions rather than entire objects.

To alleviate this problem, many methods attempt to expand the activated regions by an ensemble of different CAMs. MDC [15] noticed that dilated convolution can propagate discriminative information to surrounding regions. Therefore, it fused CAMs generated by convolutional blocks of different dilated rates to enlarge activated areas. Different from MDC, which used standard dilated convolution, FickleNet [17] employed irregular dilated convolution by applying random dropout to the hidden units. This design enabled the network to match objects of different shapes, reducing false positive regions in the combination of CAMs. OAA [23] obtained integral object regions by accumulating CAMs at different training epochs, as the classification network shifts its attention over different object parts during training. However, these ensemble approaches suffer from the problem of noise accumulation.

Another typical solution forces networks to focus on other object regions by erasing CAMs. AE-PSL [11] sequentially masked mined object regions in the input image to discover new object regions. The discovered object regions at all iterations were finally assembled to locate the entire object. Instead of partially removing the input image, ACoL [12] erased the discovered regions from feature maps. However, the erased regions of both methods consistently expand during training, leading to the overerasing problem. Redondo-Cabrera et al. [24] gave up this incremental erasing strategy and proposed H&S to seek object parts by randomly hiding patches in images. CSE [14] masked the activated regions of a random category in each multiclass image. The class-specific erasing method enlarges the captured areas while mitigating the overerasing problem by penalizing the intrusion of the CAMs.

To obtain complete pseudolabels from the generated CAMs, various methods have been developed to measure the cross-pixel affinity for postprocessing. AffinityNet [18] learns to predict a pixel-to-pixel semantic affinity matrix from the initial pseudolabels. IRN [25] produces the semantic affinity matrix based on the predicted class boundaries. As the self-attention in Transformers can build global feature relations, AFA [20] is devised to extract pixel-to-pixel semantic affinity from the Transformer. Generally, these affinity-based approaches apply random walk [26] to propagate local responses of sparse activated regions to nearby areas according to the affinity matrix. Many subsequent methods [27], [28], [29] also utilize such postprocessing to generate satisfactory pseudolabels from improved CAMs. Similarly, region growing based on low-level representations is also applied to the seed areas to extend pseudolabels [30], [31], [32].

In general, these methods search for local regions with salient patterns and aggregate them to obtain refined CAMs. Such CAMs can roughly cover entire objects with ambiguous responses falling in nonsalient or background areas. The ambiguously responding regions are commonly removed with a threshold or saliency clues [29], [33] to generate pseudolabels focusing on objects. Although CAM aggregation works for object-level targets in natural images, it fails in effectively yielding desirable CAMs for remote sensing land covers that are stuff-level with amorphous definition and large-area nonsalient material distribution. To fully capture stuff-level land covers from remote sensing images, the proposed HoLift scheme encourages each image pixel to achieve a high response in CAMs via the holistic continuity lifting loss. The guidance of this pixel-level loss function enables the classifier to expand its attention from local regions to the entire image. Consequently, our HoLiftNet can directly generate land-cover CAMs with seamless responses, eliminating the need for complex postprocessing on pseudolabels.

B. Weakly Supervised Land-Cover Classification

The success of WSSS in natural images has inspired the development of weakly supervised land-cover classification in remote sensing images. Current works mostly adopt the CAM aggregation strategy for binary weakly supervised land-cover classification, where a specific land cover is treated as a foreground object in remote sensing images. In order to capture small-scale objects in remote sensing images, WSF-Net [34] merged detailed low-level features into high-level features for CAM generation. Both SPMF-Net [35] and MSG-SR-Net [36] additionally incorporated superpixel segmentation into classification networks to localize buildings with regular shapes. Specifically, SPMF-Net employed a superpixel pooling layer to regularize the shape of incomplete and inaccurate activated regions. MSG-SR-Net fused multiscale CAMs to improve the integrity of buildings and adopted superpixels to refine building boundaries. ACGC [37] combined adversarial climbing [38] with IRN to facilitate weakly supervised building extraction. To generate pseudolabels with high integrity, UASRL [39] followed the erasing strategy of ACoL and merged the complementary CAMs in a confidence-weighted manner. These weakly supervised methods generally aggregate local responding regions to achieve holistic pseudolabels for remote sensing objects. However, as there is no foreground–background distinction in multiclass land-cover classification, the reliance on CAM aggregation can incur massive ambiguous zones into CAMs, thus severely destroying land-cover continuity in the pseudolabel.

Recently, a few multiclass WSLC methods have noticed differences between land covers and object-level targets.

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Nivaggioli and Randrianarivo [8] discovered that land covers have no background in aerial images. Motivated by this observation, the background-related designs were removed in AffinityNet to improve its performance in WSLC. Through analysis on DeepGlobe, Chan et al. [9] highlighted that class co-occurrence is challenging for WSSS. To tackle this problem, OME [40] designed a co-occurrence evaluation metric to filter out training samples. OME additionally introduced uncertainty-weighted masks to the semantic segmentation network to mitigate the interference of noise in pseudolabels. ITER [41] specifically devised spectral/spatial activation to obtain land-cover CAMs from high-dimensional hyperspectral remote sensing images. It additionally fused CAMs by multiscale flip aggregation to capture both small-scale details and large-scale patches of land covers. However, the current methods are unaware of the incompatibility between the CAM aggregation and the stuff-level characteristics of land covers.

In this article, we propose a HoLift scheme to overcome the response ambiguity in land-cover CAMs generated by the CAM aggregation strategy. Considering that the land covers are seamlessly connected, HoLift introduces a pixelwise response constraint and a global consistency constraint to generate CAMs that guarantee land-cover continuity with clear-cut borders in the pseudolabel.

III. Methodology

A. Overview

Most WSSS methods consist of two stages: 1) training a classification network to generate CAMs and 2) training a semantic segmentation network under the supervision of pixel-level pseudolabels produced from CAMs. In this study, we mainly focus on the first stage because CAM quality is critical to weakly supervised land-cover classification. In the following, we will elaborate on the HoLift scheme that eliminates ambiguous regions from land-cover CAMs by a holistic continuity lifting loss (L_{lift}) and a cross-image global consistency loss (L_{cross}). An overview of our HoLift-based weakly supervised land-cover classification network (HoLiftNet) is illustrated in Fig. 2.

During the training stage, the input to HoLiftNet is divided into input images and reference images. In order to obtain rich features from images other than the input images for enhanced semantic learning, HoLiftNet inputs multiple reference images that share at least one land-cover type with each input image. HoLiftNet outputs both CAMs and feature maps in the same manner for both input images and reference images. In detail, the backbone of HoLiftNet first encodes features from these images. The encoded features are converted into CAMs by a 1×1 convolution layer and then transformed into classification probabilities by global average pooling (GAP). In another network branch, the backbone features are fed into a multilayer perceptron (MLP) to yield projected feature maps. We compute L_{lift} based on CAMs generated in the feed-forward pass to force seamless area responses in CAMs. L_{cross} is calculated between projected feature maps of different images to enhance the global feature consistency of each category.

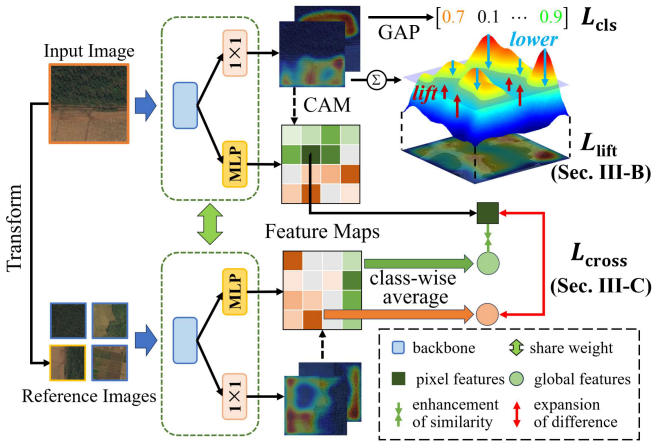


Fig. 2. Overview of the image classification network built with our HoLift scheme. The classifier extracts CAMs and classification probabilities of the input image. The classification and CAM generation are guided by the classification loss L_{cls} and the holistic continuity lifting loss L_{lift}, respectively. The network also extracts classwise average features as global features from each reference image to enhance the pixel features of the input image. The cross-image global consistency constraint L_{cross} increases intraclass consistency (green arrows) and interclass discrimination (red arrows) between pixel features and global features. GAP denotes global average pooling. MLP refers to the projection head [42] built with a multilayer perceptron.

We first briefly introduce CAM generation and pixelwise classification before detailing the proposed L_{lift} and L_{cross}. Following [12] and [16], we adopt VGG-16 [43] with three additional convolutional layers as our backbone. The backbone feature maps are mapped into CAMs via a 1×1 convolution, as illustrated in Fig. 2. We denote CAMs of all C classes as $\mathbf{M} \in \mathbb{R}^{C \times H \times W}$, where $H \times W$ is the spatial size. By applying three max pooling layers in VGG-16, the output CAMs are eight times smaller than the input image in both width and height.

Subsequently, the category of each pixel can be determined by comparing activation values between different classes. Each multiclass CAM $\mathbf{M}$ is first processed by a ReLU layer and then normalized to $[0, 1]$ by the classwise maximum values

$$\bar{M}_c = \frac{\text{ReLU}(M_c)}{\max(M_c)}, \quad 1 \le c \le C \tag{1}$$

where M_c is the CAM of the cth class. Upon the normalized CAMs, i.e., $\bar{\mathbf{M}}$, we can assign a category for each pixel by comparing the activation values between classes

$$P_{i,j} = \begin{cases} \text{argmax}\left(\bar{M}_{\cdot,i,j}\right), & \max\left(\bar{M}_{\cdot,i,j}\right) \ge t \\ \text{Invalid class}, & \max\left(\bar{M}_{\cdot,i,j}\right) < t \end{cases} \tag{2}$$

where $\bar{M}_{\cdot,i,j}$ refers to the activation values of pixel (i, j) in the normalized CAMs of all labeled classes, $\text{argmax}(\bar{M}_{\cdot,i,j})$ outputs the class with the highest activation value, and t is a threshold value to exclude low-response pixels that are uncertain in determining the category. By bilinearly upsampling the normalized CAMs to the size of the input image, we can obtain pixel-level pseudolabels of the input image using (2).

B. Holistic Continuity Lifting Loss

In the first stage of WSSS, the image classifier only focuses on the most discriminative regions for classification,

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resulting in CAMs that partially cover objects [44]. Previous methods mainly aggregate multiple discriminative regions to expand responding areas for an entire object. Despite being effective for natural images, it usually fails to completely capture the stuff-level land covers in remote sensing images, which are amorphous without salient patterns. The incomplete responding regions could bring discontinuity to pseudolabels. As remote sensing images are seamlessly covered by land covers, each pixel needs to be activated to obtain holistic pseudolabels. Ideally, each pixel gets activated in the CAM of only one category to avoid semantic confusion.

In order to obtain CAMs with seamless responding areas, we design a holistic continuity lifting loss (L_{lift}), which imposes constraints on CAM pixels to encourage a high activation value for each pixel in a specific category. Numerically, each pixel in an ideal normalized CAM ($\bar{\mathbf{M}}$) has an activation value of 1 in the correct category and 0 in the other categories. However, the image-level labels for weakly supervised learning cannot provide pixelwise ground truth. Therefore, we expect the pixelwise sum of activation values of normalized CAMs to be 1. This expectation encourages a high response for each CAM pixel and prevents confusion between classes. Furthermore, the categories absent in each image should be excluded from the summation so that the network can focus on the labeled classes of the image. Based on the above consideration, L_{lift} is defined as follows:

$$L_{\text{lift}} = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} \left(\sum_{c \in C_k} \bar{M}_{c,h,w} - 1 \right)^2 \tag{3}$$

where C_k represents the labeled classes of an image, and $\bar{M}_{c,h,w}$ is the activation value of pixel (h, w) in $\bar{\mathbf{M}}$ of the cth category.

As shown in (3), L_{lift} sums the normalized CAMs of the labeled image classes pixel by pixel and encourages the sum of each pixel to approach 1. On the one hand, in low-response regions that break the continuity of pseudolabels (i.e., sum < 1), L_{lift} lifts activation values to expand activated regions. On the other hand, in the ambiguous regions with confusion between categories (i.e., sum > 1), L_{lift} decreases activation values to suppress CAM noise. By imposing pixel-level constraints on CAMs, L_{lift} forces the network to extend its attention from local discriminative regions to the entire image, thus alleviating semantic ambiguity in CAMs.

Additionally, L_{lift} can amplify the distinction between high and low responses in CAMs of multiclass images for better land-cover discrimination. For the single-class images with only one labeled class, L_{lift} tends to increase activation values of the labeled class throughout the entire image. As for multiclass images, if a pixel achieves one positive activation value exclusively for a single class, L_{lift} will lift that value, as the ReLU function used in (1) will eliminate the gradients of the other categories. Consequently, other low-response regions resembling areas with enhanced CAM responses can also attain high activation values for the specific class. Meanwhile, L_{lift} reduces the responses of the remaining classes by decreasing the sum greater than 1. In other words, L_{lift} directs the responses toward 1 and 0. As a result, the ambiguously responding regions in CAMs are reduced, making it easier to distinguish between different land-cover types. In brief, the training guidance of L_{lift} facilitates the generation of high-quality CAMs that holistically describe land covers with few interclass conflicts.

C. Cross-Image Global Consistency Loss

Although L_{lift} can significantly improve the completeness of pseudolabels, it may bring semantic errors into the produced pseudolabels. Due to the absence of fine-grained semantic guidance, activation values of incorrect classes may also be lifted by L_{lift}, especially at easily confused pixels around land-cover boundaries, leading to inaccurate pseudolabels. To improve the reliability of L_{lift} in multiclass images and obtain CAMs with accurate boundaries, we further introduce a cross-image global consistency loss (L_{cross}) to enhance the learning of semantic information.

WSSS methods usually strengthen the learning of semantic information by establishing regional semantic relations within natural images [18], [20]. For objects with regular shapes and constant components in natural images, building intraimage relations between object parts helps networks learn global object patterns, which is applicable across images. However, the intraimage region relations are less effective for stuff-level land covers without representative components and well-defined shapes. Due to the diverse geographic distributions of remote sensing images, land cover exhibits various appearances across images. Therefore, it is essential to construct regional semantic relations between images to enable networks to discover global land-cover patterns from diverse appearances.

To this end, we explore regional relations across images and strengthen the semantic learning of our HoLiftNet by introducing L_{cross}. L_{cross} is devised to increase global feature consistency between cross-image regions of the same class and improve interclass feature differences between land-cover regions across images. The regional categories are predicted from the generated CAMs using (2), and pixels predicted as the invalid class are excluded from the calculation of L_{cross}. Inspired by contrastive learning, we develop L_{cross} upon the InfoNCE loss [45], [46]. Given an pair of input image q and reference image k that share at least one class, L_{cross} can be calculated by

$$L_{\text{cross}}(q,k) = \frac{L_{\text{NCE}}(q,k) + L_{\text{NCE}}(k,q)}{2}, \tag{4}$$

$$L_{\text{NCE}}(q,k) = -\frac{1}{|V_q|} \sum_{i \in V_q} \log \frac{\exp\left(F_i^q \cdot \overline{F_+^k}/\tau\right)}{\sum_{c \in C_k} \exp\left(F_i^q \cdot \overline{F_c^k}/\tau\right)} \tag{5}$$

where V_q represents pixels with one valid class (i.e., with a maximum activation value greater than the threshold t) in the CAM of image q, $|V_q|$ is the number of these pixels, F_i^q denotes the feature vector of pixel i in image q, $\overline{F_c^k}$ refers to the average feature vector of class c in image k, and $\overline{F_+^k}$ is the average feature vector of pixel i's category in image k. The category of each pixel is determined by (2). In (5), the feature vectors are normalized

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by $L2$-norm [47], which ensures that the dot product between feature vectors represents the cosine similarity. We regard the classwise average feature vector $\overline{F_c^k}$ as the optimization reference of F_i^q. Therefore, we ignore the gradient of $\overline{F_c^k}$ during the training process to avoid simultaneously updating F_i^q and $\overline{F_c^k}$. τ is a temperature hyperparameter [47].

As shown in Fig. 2 and (5), L_{cross} first computes classwise average feature vectors from the projected feature maps of a reference image k. Subsequently, classwise cosine similarity values are calculated between the feature vector of each pixel in the input image q and the average feature vectors of all classes in image k. In (5), the intraclass cosine similarity value is divided by the sum of cosine similarity values across all categories. Minimizing $L_{\text{NCE}}(q,k)$ aligns the features in image q with those of homogeneous regions in image k, contributing to higher similarity within the same class and greater differences between classes. By exchanging the roles of q and k in (4), L_{cross} simultaneously optimizes the features of both images.

In order to optimize global semantic features more effectively, our HoLiftNet inputs N reference images $k_i (i = 1, 2, \ldots, N)$ for each input image q to increase the diversity of classwise average features. In particular, the last reference image k_N is transformed from q to make use of intraimage semantic relations. As shown in Fig. 2, our HoLiftNet adopts a projection head to map backbone feature maps into 256 channels for feature optimization. The projection head consists of a two-layer MLP, which has been demonstrated to be effective in improving the features of the preceding layers during contrastive learning [42]. The projection head is only applied during training and is discarded during inference. The whole L_{cross} is defined by

$$L_{\text{cross}} = \frac{1}{N} \sum_{i}^{N} L_{\text{cross}}(q, k_i). \tag{6}$$

D. Overall Loss

Our HoLiftNet obtains classification probabilities of images by feeding CAMs into a GAP layer. The probability of each class is calculated by

$$p_c = \sigma\left[\text{GAP}\left(M_c\right)\right], \quad 1 \le c \le C \tag{7}$$

where $\sigma(\cdot)$ is the sigmoid function. The classification loss L_{cls} is defined as follows:

$$L_{\text{cls}} = -\frac{1}{C} \sum_{c=1}^{C} \left[y_c^q \times \log p_c^q + \left(1 - y_c^q\right) \times \log\left(1 - p_c^q\right) \right] \tag{8}$$

where $y_c^q \in \{0, 1\}$ indicates whether class c exists in the input image q, and p_c^q is the predicted probability of class c existing in image q.

To prevent overfitting, we only compute L_{cls} and L_{lift} for image q. The overall loss for training HoLiftNet is defined as follows:

$$L = L_{\text{cls}} + \lambda_1 L_{\text{lift}} + \lambda_2 L_{\text{cross}} \tag{9}$$

where weights λ_1 and λ_2 are set to 1 and 4 in practice. This choice was made to roughly balance the magnitudes of all loss terms. Further experiments were conducted by varying both weights to search for better configurations. The performance remains stable within a reasonable range of both weights, and the setting $\lambda_1 = 1$ and $\lambda_2 = 4$ yielded the best overall performance.

Algorithm 1 HoLift Scheme in a PyTorch-Like Style

```
# q, k: input image and reference image
# l_q, l_k: label of q and k in one-hot format
# prepare N reference images
[k_1, ..., k_{N-1}] = select_reference(q) # each k shares a category with q and at least one contains multiple categories
K = [k_1, ..., k_{N-1}, aug(q)] # add augmented q to K
# extract backbone features
F_q = Backbone(q) # shape: D × W × H
F_k = Backbone(k) # extract feature for each k in K
# generate CAMs
CAM_q = CAM_Head(F_q) # shape: C × W × H
CAM_k = CAM_Head(F_k)
# holistic continuity lifting loss
L_lift = CAM_q[l_q, :, :].sum(0).mean() # pixel-wise sum across classes
# predict image classes
p_q = GAP(CAM_q) # classification score, shape: C
L_cls = L_CE(p_q, l_q) # compute cross entropy loss
# cross-image global consistency loss
P_q = CAM_q.argmax(0) # generate pseudo labels
P_k = CAM_k.argmax(0) # shape: 1 × H × W
mask_q = one_hot(P_q) ⊙ (CAM_q.max(0) > threshold) ⊙ l_q
# get masks recording predicted regions of each category
mask_k = one_hot(P_k) ⊙ (CAM_k.max(0) > threshold) ⊙ l_k
# shape: C × H × W
F'_q = MLP(F_q) # get projected feature maps
F'_k = MLP(F_k) # shape: D' × H × W
F'_q = mm(mask_q.view(C, -1), F'_q.view(-1, D'))/mask_q.sum(1, 2) # class-wise average features
F'_k = mm(mask_k.view(C, -1), F'_k.view(-1, D'))/mask_k.sum(1, 2) # shape: C × D'
sim_qk = mm(norm(F'_q, 0).view(HW, D'), norm(F'_k, 1).view(D', C)) # cosine similarity between pixel features and class average features
sim_kq = mm(norm(F'_k, 0).view(HW, D'), norm(F'_q, 1).view(D', C)) # shape: HW × C
L_cross = Σ_{i=1}^{N} (L_CE(sim_{qk_i}, P_q) + L_CE(sim_{k_i q}, P_{k_i}))/2N
# compute overall loss
L = L_cls + λ_1 L_lift + λ_2 L_cross
L.backward()
```

⊙: Hadamard product; mm: matrix multiplication; norm: $L2$-norm.

The overall pseudocode of HoLift scheme is shown in Algorithm 1.

IV. Experiments

To validate the effectiveness of HoLift scheme for weakly supervised land-cover classification, we evaluate its performance on two challenging remote sensing datasets, including DeepGlobe Land Cover Classification Challenge [48]

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and 2020 IEEE GRSS DATA FUSION CONTEST (DFC2020) [49], [50]. In this section, we will detail the two datasets, experimental settings, results, and comprehensive analysis.

A. Dataset

DeepGlobe [48] contains 803 high-resolution RGB images with a size of 2448 × 2448. We use the commonly used training/validation/testing settings [51] and crop each image into 256 × 256 pixels without overlap. The dataset is well annotated with six classes, namely urban land, agricultural land, rangeland, forest land, water, and barren land.

DFC2020 [49], [50] consists of 85382/986/5128 satellite images for training/validation/testing, each image with a size of 256×256. Each image contains two SAR bands acquired by Sentinel-1 and 13 spectral bands acquired by Sentinel-2. For land-cover classification, we discard three atmosphere-related spectral bands of images following [52]. We adopt the preprocess commonly used in previous studies [52], [53] for the DFC2020 dataset. Specifically, the pixel values of the Sentinel-1 SAR bands are clipped to the range of $[-25, 0]$, while those of the Sentinel-2 optical bands are truncated to $[0, 10000]$. Both modalities are then normalized to the range $[0, 1]$. During training, the SAR bands are treated as an additional spectral channel and stacked with the optical images. DFC2020 provides pixel-level annotations for each image. The validation and testing sets are finely annotated while the training labels derived from SEN12MS [54] are relatively coarse. The image pixels are classified into ten categories: forest, shrubland, savanna, grasslands, wetlands, croplands, urban, snow, barren, and water. Since two geographically specific categories, i.e., savanna and snow, are absent in the validation and testing subsets, we exclude images containing these two classes from network training to ensure that both training and evaluation only involve the remaining eight classes, following existing methods [55].

Although both datasets provide pixel-level annotations in training subsets, our training only uses image-level labels for weakly supervised land-cover classification. In addition, we adopt the high-resolution training annotations of DeepGlobe to evaluate the generated CAMs.

B. Experimental Setup

1) Implementation Details: The proposed HoLiftNet is implemented with PyTorch and all image-level classifiers are trained with 4 NVIDIA GTX 2080Ti GPUs with a batch size of 64. Learning rates are initialized as 1e-3 and 5e-4 for DeepGlobe and DFC2020, respectively, and updated by a polynomial learning schedule. The training lasts 30 epochs with an SGD optimizer that has a momentum of 0.9 and a weight decay of 5e-4. The first two training epochs only involve L_{cls}, enabling HoLiftNet to roughly identify land covers. Then, our L_{lift} and L_{cross} join in training after two epochs. The number N of reference images is 4, and we use random resizing, random flipping, and random rotating (90°) to generate the last reference image (k_N). The threshold t in (2) is set to 0.7 during training in order to extract representative features for contrastive learning, and it is reduced to 0.5 during evaluation and inference to obtain more holistic pseudolabels. The temperature hyperparameter τ in L_{cross} [see (5)] is set to 0.1.

In the second phase of weakly supervised land-cover classification, we employ DeepLabv3+ [56] for pixel-level land-cover classification. We train it with pixel-level pseudolabels produced in the first stage, and the training batch size is 192. We warm up the network for three epochs and the learning rate drops from $3e^{-4}$ by a polynomial learning schedule in the next 27 epochs. The other hyperparameters are the same as the first stage.

2) Evaluation Metrics: We use mean intersection over union (mIoU) and average accuracy (AA) to evaluate the land-cover classification performance on testing sets of DeepGlobe and DFC2020 datasets, respectively. In addition, we adopt recall and noise rate [23] to assess the quality of CAMs on the training set of DeepGlobe. The metrics are formulated as follows:

$$\text{mIoU} = \frac{1}{C}\sum_{i=1}^{C}\frac{\text{Pred}_i \bigcap \text{GT}_i}{\text{Pred}_i \bigcup \text{GT}_i} \tag{10}$$

$$\text{AA} = \frac{1}{C}\sum_{i=1}^{C}\frac{\text{Pred}_i \bigcap \text{GT}_i}{\text{GT}_i} \tag{11}$$

$$\text{recall} = \frac{\sum_{i=1}^{C}\left(\text{AT}_i \bigcap \text{GT}_i\right)}{\sum_{i=1}^{C}\text{GT}_i} \tag{12}$$

$$\text{noise rate} = 1 - \frac{\sum_{i=1}^{C}\left(\text{AT}_i \bigcap \text{GT}_i\right)}{\sum_{i=1}^{C}\text{AT}_i} \tag{13}$$

where C is the class number, Pred_i refers to the pixels predicted as class i, GT_i represents the pixels labeled as category i, and AT_i means the pixels with activation values higher than the threshold t in the CAM of class i. The recall indicates the proportion of pixels with high activation values, while the noise rate reveals the proportion of pixels with semantic confusion in CAMs. Both the recall and noise rate of CAMs are computed according to the ground truth of the training subset. As introduced in Section IV-A, DeepGlobe is precisely labeled while the training set of DFC2020 only provides low-resolution annotations. Therefore, we only evaluate the quality of CAMs on DeepGlobe.

C. Results on DeepGlobe Dataset

To demonstrate the effectiveness of our HoLift scheme in generating reliable land-cover CAMs and thereby improving weakly supervised land-cover classification, we compare the performance of our HoLiftNet with state-of-the-art methods on the DeepGlobe dataset. For a comprehensive evaluation, HoLift scheme is implemented with two commonly used backbones, i.e., VGG-16 with three additional convolutional layers [12] and ResNet-50 [57]. The quantitative results are displayed in Table I.

As shown in Table I, the pioneer VGG-GAP delivers both low recall and noise rates, indicating that it only captures small land-cover areas in remote sensing images. The successors

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TABLE I

QUANTITATIVE RESULTS ON DEEPGLOBE DATASET. THE RECALL AND NOISE RATES OF AFFINITYNET ARE IGNORED AS IT DIRECTLY GENERATES REFINED PSEUDOLABELS FROM THE INITIAL CAMS OF VGG-GAP. † MEANS THE POSTPROCESSING OF IRN IS EMPLOYED TO REFINE PSEUDOLABELS. FOR MULTISTAGE METHODS, TIME CONSUMPTION AND PARAMETER SIZE ARE REPORTED AS THE CUMULATIVE TOTALS ACROSS ALL STAGES. TRAINING AND INFERENCE TIME ARE COUNTED ON FOUR GPUS AND ONE GPU, RESPECTIVELY. THE BEST ARE GIVEN IN **BOLD**. ↑: HIGHER IS BETTER, ↓: LOWER IS BETTER

Method	Backbone	mIoU↑	noise↓	recall↑	training time↓	inference time↓	parameters↓
VGG-GAP	VGG-16	45.98	**21.33**	20.06	**1h41min**	24min	21.80M
RCA	VGG-16	47.61	30.02	54.78	16h07min	**17min**	70.57M
DRS	VGG-16	49.29	28.93	59.50	2h42min	26min	24.39M
MCIS	VGG-16	50.22	22.62	40.51	7h12min	1h42min	22.33M
AffinityNet	VGG-16	51.32	-	-	3h16min	32min	**21.05M**
Puzzle-CAM	VGG-16	51.66	31.03	59.31	2h24min	24min	21.80M
OAA	VGG-16	52.56	24.28	52.95	1h52min	**17min**	21.80M
HoLiftNet(ours)	VGG-16	**53.91**	23.30	**75.58**	11h40min	24min	22.19M
LPCAM	ResNet-50	45.25†	24.35	44.97	2h33min	7h32min	48.42M
SFC	ResNet-50	50.64†	31.25	53.64	**1h42min**	6h56min	53.51M
ReCAM	ResNet-50	52.68†	31.71	91.34	4h33min	6h32min	71.95M
IRN	ResNet-50	53.25†	22.76	52.27	2h33min	5h53min	48.42M
AdvCAM	ResNet-50	53.59†	33.38	**98.42**	2h33min	5h58min	48.42M
HoLiftNet(ours)	ResNet-50	**54.89**	**21.02**	76.50	5h45min	**26min**	**24.70M**

proposed various solutions to expand the areas with high activation values, which significantly improved the CAM recall. However, the recall rate of CAMs generated by these methods tailored for natural images remains around 50%. Meanwhile, some methods, such as RCA and Puzzle-CAM, fail to suppress the activation values of confusing categories, which introduce severe noise to CAMs. As a result, the produced pseudolabels with low coverage and strong noise limit the accuracy of land-cover classification. In contrast, our HoLiftNet with a VGG-16 backbone significantly outperforms other methods by at least 16.08% in terms of recall. HoLiftNet takes the stuff-level characteristics of remote sensing land covers into account, thus effectively enlarging the activation areas by encouraging seamless area response in CAMs. In addition to high coverage, CAMs generated by HoLiftNet achieve a low noise rate of 23.30%. Some methods achieve a lower noise rate at the great expense of recall. However, we can obtain CAMs with a low noise rate without sacrificing responding areas. The superiority of HoLiftNet indicates its strong capability of distinguishing between different land-cover classes, which overcomes the challenge of high intraclass similarity in remote sensing images. By generating pseudolabels from such high-quality CAMs, the land-cover classification model yields the top mIoU score of 53.91% among comparison methods.

In general, classifiers with the ResNet backbone tend to perform better than VGG-based methods in land-cover classification, due to its stronger deep feature extraction ability. It is interesting that our VGG-based HoLiftNet outperforms the ResNet-based competitors in mIoU. Furthermore, by replacing the backbone with ResNet-50, HoLiftNet achieves a high recall of 76.50% and the lowest noise of 21.02%. Although ReCAM and AdvCAM significantly lift CAM responses and achieve extremely high recall scores, they suffer from severe interclass uncertainty, which results in noise rates higher than 30%. The noisy CAMs could lead to unsatisfactory pseudolabels that severely limit the performance of land-cover classification. Unlike these comparison methods, the high-quality CAMs of HoLiftNet demonstrate its strong capability of holistic land-cover recognition and interclass discrimination. Benefiting from the stronger backbone, the mIoU of HoLiftNet increases by 0.98%, reaching 54.89%, which outperforms state-of-the-art methods by 1.30% without the use of extra postprocessing.

In terms of efficiency and resource usage, HoLiftNet requires more training time than most single-image methods, but maintains a short inference time and a small parameter size. This is because the two proposed losses are designed to fully exploit the full-coverage priority and cross-image regional relations of land covers, while avoiding complex network modifications to ensure high scalability. We argue that the additional several hours of training are acceptable given the significant accuracy gains and the substantial time and labor saved by avoiding pixel-level annotations required in fully supervised learning. Compared to other cross-image methods such as RCA (16 h 07 min training time, 70.57 M parameters) and MCIS (1 h 42 min inference time), HoLiftNet achieves better accuracy with comparable or lower resource consumption. Moreover, compared to advanced ResNet-50 methods, HoLiftNet attains the highest mIoU while drastically reducing inference time (26 min versus 6–7 h) and model size (24.70 M versus 48–72 M). This advantage comes from its single-stage pseudolabel generation without complex postprocessing, which makes HoLiftNet more practical and efficient for large-scale land-cover mapping. Fig. 3 illustrates the CAMs generated by various VGG-based methods for visual inspection.

In Fig. 3, we select both single-class and multiclass images to investigate the performance of weakly supervised

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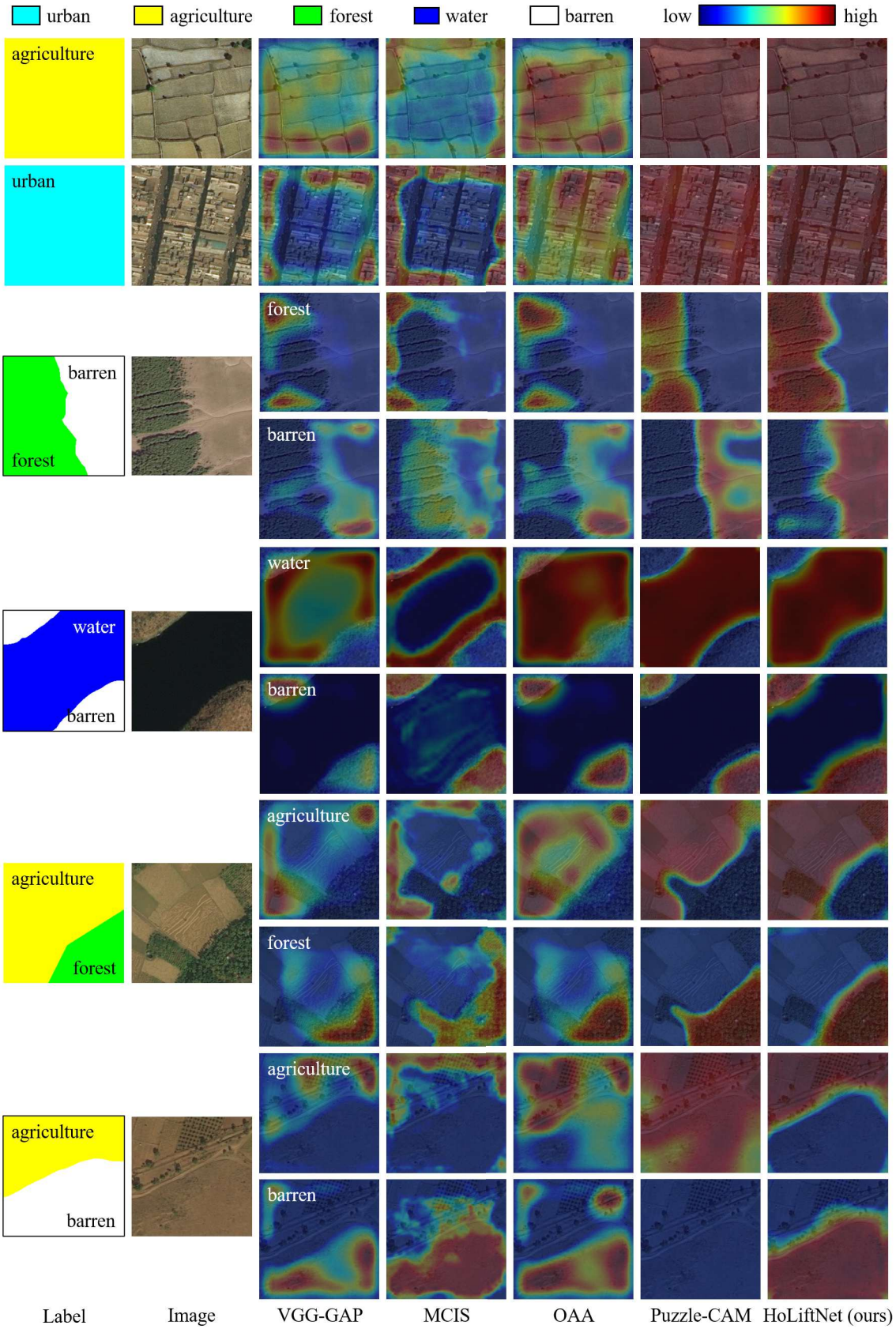


Fig. 3. Qualitative comparison on DeepGlobe dataset.

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TABLE II
QUANTITATIVE RESULTS ON DFC2020 DATASET

Method	AA(%) ↑
VGG-GAP [10]	40.67
ReCAM [58]	35.28
AdvCAM [38]	39.35
RCA [59]	39.60
IRN [25]	39.95
SEAM [60]	39.97
MCIS [61]	40.25
L2G [62]	40.27
OAA [16]	40.35
Puzzle-CAM [63]	40.85
HoLiftNet(ours)	**41.87**

approaches. As shown in Fig. 3, VGG-GAP can only capture sparse land-cover regions in both types of remote sensing images. Although OAA can enlarge high-response regions by aggregating CAMs of various epochs, the problem of partial activation remains unsolved. Additionally, CAM noise from different epochs propagates to the final CAMs, leading to high CAM responses in the false category, as illustrated in the fifth column of Fig. 3. The visualization results also indicate the weaknesses of MCIS in completely and accurately locating land covers. In terms of single-class images, both Puzzle-CAM and our HoLiftNet produce perfect CAMs with full coverage. However, unactivated regions can also be observed in the CAMs of Puzzle-CAM in the first multiclass image. Moreover, Puzzle-CAM confuses the green agricultural areas with forest areas of a similar color in the multiclass image of the fifth row in Fig. 3. In the last multiclass image with higher inter-class similarity, Puzzle-CAM misclassifies all barren land as agricultural land. In contrast, our HoLiftNet generates more continuous and accurate CAMs for multiclass images. Benefiting from the enhancement of interclass feature differences achieved by the cross-image global consistency loss (L_{cross}), our HoLiftNet clearly distinguishes agricultural lands from forest areas according to the internal regularity of the plots. The visual results demonstrate that our HoLift scheme is effective in holistically recognizing land covers and clearly distinguishing between classes.

D. Results on DFC2020 Dataset

For a comprehensive evaluation, we additionally carry out experiments on the challenging DFC2020 dataset. Different from DeepGlobe, DFC2020 only provides low-resolution annotations for the training samples. Although our HoLiftNet only requires image-level supervision, these coarse labels are inaccurate for the evaluation of CAMs generated from training images. Therefore, we only use the official AA metric for performance assessment of land-cover classification on the testing subset. The numerical results of HoLiftNet on DFC2020 are reported in Table II.

As shown in Table II, our VGG-based HoLiftNet achieves a top AA of 41.87%, which outperforms state-of-the-art methods by 1.02%. The results again manifest the effectiveness of HoLift in weakly supervised land-cover classification. One may notice that the vanilla VGG-GAP yields a high AA of 40.67% on DFC2020, excelling most of its competitors. We speculate that the difference in land-cover distribution between DFC2020 and DeepGlobe leads to the performance change of VGG-GAP. The image resolution of DFC2020 is lower than that of DeepGlobe, meaning that each DFC2020 image records a larger land surface. Small-area land-cover patches become more common in DFC2020 images, and the intraclass appearance diversity increases within images. When generating CAMs for such complex images, VGG-GAP can conservatively capture the most discriminative regions with minimal noise, contributing to the high land-cover classification accuracy in the second stage. Some qualitative results on DFC2020 are displayed in Fig. 4.

TABLE III
ABLATION STUDY ON DEEPGLOBE DATASET. THE BEST ARE GIVEN IN **BOLD**. ↑: HIGHER IS BETTER, ↓: LOWER IS BETTER

Method	L_{lift}	L_{cross}	mIoU ↑	noise ↓	recall ↑
Baseline			45.98	**21.33**	20.06
HoLiftNet	✓		52.76	23.91	75.48
HoLiftNet		✓	35.16	25.52	13.52
HoLiftNet	✓	✓	**53.91**	23.30	**75.58**

Fig. 4 compares the CAMs produced by various weakly supervised land-cover approaches on DFC2020. The single-class image depicted in the first row of Fig. 4 further demonstrates the superiority of HoLiftNet in holistically responding land covers. As for multiclass images, CAMs of VGG-GAP roughly cover images with high precision. Unfortunately, ambiguous regions still exist in the CAMs of the last image. OAA enlarges activated areas, but mainly in CAMs of incorrect classes. Although Puzzle-CAM refines the CAMs, it still generates regions with low activation values (e.g., croplands in the first multiclass image) or with high responses on false categories (e.g., urban areas on the left in the second multiclass image). In contrast, our HoLift scheme effectively mitigates these problems.

E. Ablation Studies

1) HoLift Scheme: Ablation studies were conducted for our HoLift scheme on the DeepGlobe dataset. VGG-GAP [10] trained with the classification loss L_{cls} is chosen as the baseline. We first investigate the effect of the holistic continuity lifting loss (L_{lift}) and the cross-image global consistency loss (L_{cross}) independently. Table III lists the quantitative results of models trained by different settings of the loss function.

As illustrated in Table III, the baseline generates inferior CAMs and limits the accuracy of land-cover classification. The application of L_{lift} contributes to a significant increase in both recall and mIoU, which indicates that L_{lift} is effective in lifting CAM responses. However, the noise rate also increases with the expansion of responding areas. The precision degradation of CAMs enlarges misclassified regions in the generated pseudolabels, which mislead the learning of land-cover classification models. The increase in noise is due to the lack of semantic guidance in L_{lift}. Since L_{lift} imposes constraints on activation values without differentiating among different types of land cover, it is unable to prevent activation values from being lifted on the wrong category. As shown in the last row of

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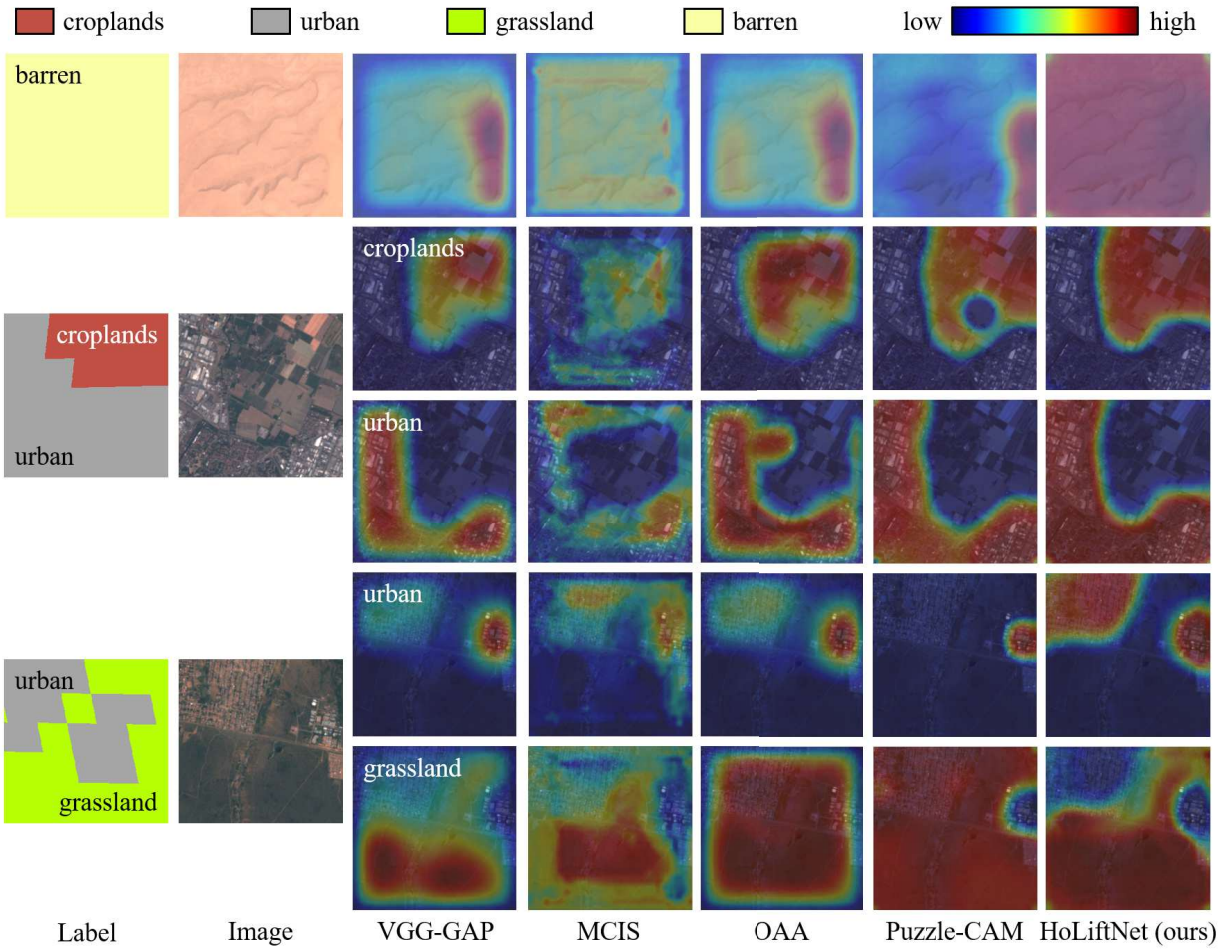


Fig. 4. Qualitative comparison on DFC2020 dataset. Note that the resolution of pixel-level labels in the training set is too low to reflect precise locations of land covers. Semantic edges of ground truth can be discerned from texture differences in images.

Table III, incorporating L_{cross} into the loss function reduces the CAM noise rate, thereby boosting the mIoU score by 1.15%. Such improvement indicates that L_{cross} alleviates the problem caused by L_{lift} by establishing regional semantic correlations between images and optimizing the global feature distribution. However, Table III shows that only combining L_{cross} with L_{cls} severely impairs the CAM quality. The possible reason is that the training is trapped in a local optima, where L_{cross} decreases its value by shrinking activated regions to those with the highest intraclass similarity. Such CAM degradation can be prevented by L_{lift}, which encourages larger activated regions. In brief, L_{lift} and L_{cross} can complement each other and cooperate to achieve the best performance. The visualization results are presented in Fig. 5 for a perceptual inspection of the effect of L_{lift} and L_{cross}.

As shown in Fig. 5, CAMs of the baseline only cover sparse land-cover regions. Under the guidance of L_{lift}, the regions with high responses significantly expand in CAMs of various land-cover classes, forming a full coverage of each image. The reduction of ambiguous regions demonstrates that L_{lift} holistically lifts responses in CAMs and thus enhances the continuity of the produced pseudolabels. Meanwhile, the responding areas are complementary in CAMs of different classes. The significant interclass differences in CAM responses indicate that semantic ambiguity is alleviated. The improvement can be attributed to the use of L_{lift}, which effectively lifts single-class responses in CAMs while suppressing high activation values across multiple categories. However, some areas near semantic boundaries are highlighted in CAMs of false classes, as depicted in the red rectangles of Fig. 5. The misclassification problem is well alleviated by L_{cross} that enhances semantic guidance. As shown in the last two columns of Fig. 5, the use of L_{cross} improves the localization precision of land covers around semantic edges and reduces ambiguously responding regions to produce clear-cut boundaries. In the following, we further study different settings of L_{lift} and L_{cross}, respectively.

2) Holistic Continuity Lifting Loss: Our L_{lift} can be divided into two parts: 1) increasing activation values for pixels with a response sum lower than 1 and 2) decreasing activation values for pixels with a response sum higher than 1. Table IV illustrates the quantitative analysis for both parts of L_{lift}.

As shown in Table IV, calculating L_{lift} only for low-response pixels yields a top recall of up to 86.90%. The significant improvement indicates that L_{lift} can enlarge activated areas by lifting activation values. However, the noise rate also increases by 7.96%. This is because the enhancement of activation values lacks constraints. The activation values in easily confused regions often simultaneously increase on multiple classes,

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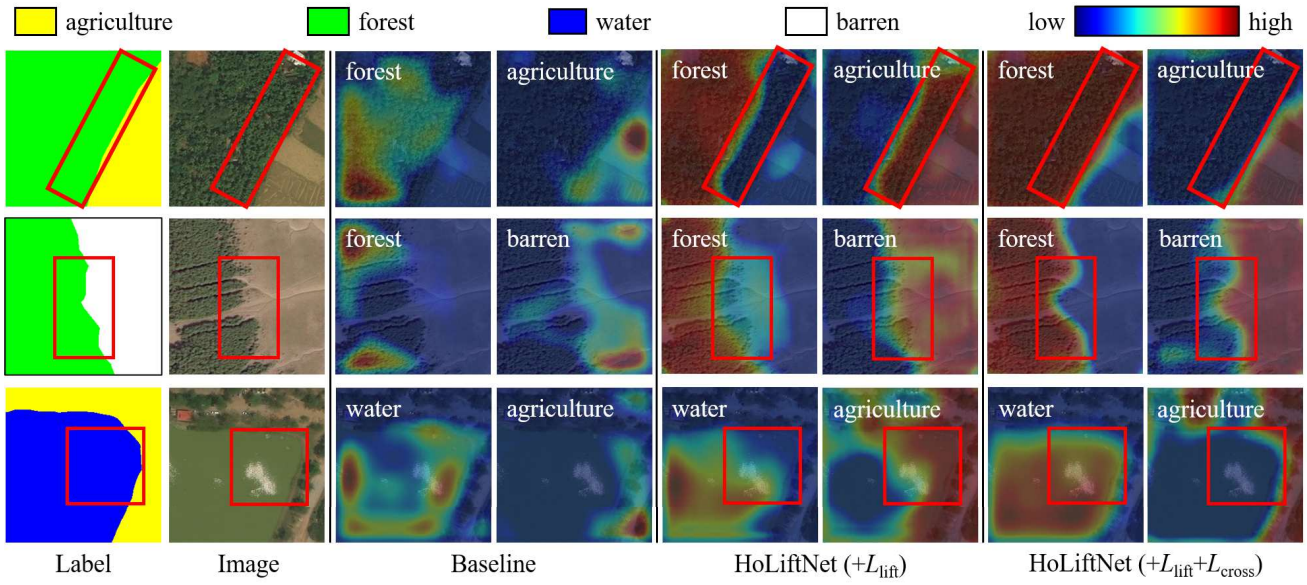


Fig. 5. Qualitative results of the ablation study.

TABLE IV

ABLATION STUDY FOR L_{LIFT}. sum < 1 MEANS WE ONLY COMPUTE L_{LIFT} FOR PIXELS WHOSE SUM OF ACTIVATION VALUES ACROSS ALL LABELED CATEGORIES IS LOWER THAN 1

Method	sum < 1	sum > 1	mIoU↑	noise↓	recall↑
Baseline			45.98	21.33	20.06
HoLiftNet	✓		52.60	29.29	**86.90**
HoLiftNet		✓	44.65	**20.81**	19.36
HoLiftNet	✓	✓	**52.76**	23.91	75.48

TABLE V

ABLATION STUDY ON DEEPGLOBE DATASET FOR $L_{\text{CROSS}}(q,k)$. (k,q) MEANS THAT $L_{\text{NCE}}(k,q)$ SHOWN IN (6) IS CALCULATED IN $L_{\text{CROSS}}(q,k)$. T_q SHOWS WHETHER IMAGE k_N IS TRANSFORMED FROM IMAGE q

T_q	$L_{\text{cross}}(q,k)$				mIoU↑ (%)	noise↓ (%)	recall↑ (%)
	(q,k)	(k,q)	(q,q)	(k,k)			
					52.76	23.91	75.48
✓	✓				53.77	**23.27**	75.27
✓	✓	✓			**53.91**	23.30	**75.58**
	✓	✓			53.69	23.40	75.18
	✓	✓	✓		53.31	23.83	74.38
	✓	✓	✓	✓	52.20	24.49	73.49

increasing semantic uncertainty. Applying L_{lift} to pixels with response sums greater than 1 can effectively reduce noise by suppressing responses in easily confused regions. The complete L_{lift} achieves a balance between recall and precision, and eliminates ambiguous regions in CAMs to the maximum extent, contributing to the highest mIoU of 52.76%.

3) Cross-Image Global Consistency Loss: As shown in Table III, adopting L_{cross} alone for training impairs the CAM quality. Therefore, the entire loss function incorporates L_{lift} for training when conducting ablation studies on L_{cross}. According to (6), L_{cross} is mainly influenced by $L_{\text{cross}}(q,k)$ and the number of reference images. We first quantitatively compare different calculation methods of $L_{\text{cross}}(q,k)$ in Table V. Given an input image q and a reference image k, the simplest $L_{\text{cross}}(q,k)$ only contains $L_{\text{NCE}}(q,k)$ that optimize features of q based on classwise average features of k. The first two rows of Table V show that this loss reduces CAM noise. Additionally, we can exchange the role of images q and k to form $L_{\text{NCE}}(k,q)$ and add it to $L_{\text{cross}}(q,k)$. As illustrated in Table V, such modification achieves a similar low noise rate and performs better in recall and mIoU scores. The improvement demonstrates that simultaneously optimizing features of both images is more effective than only optimizing features of image q. In L_{cross}, we specifically transform the input image q as the last reference image k_N to take advantage of regional relationships within each image. As shown in the third row of Table V, replacing the transformed q with another reference image leads to a slight degradation in the performance of CAMs, demonstrating the importance of intraimage regional relations in improving global feature consistency. To enhance feature distribution within each image, we also attempt to introduce $L_{\text{NCE}}(q,q)$ and $L_{\text{NCE}}(k,k)$ to $L_{\text{cross}}(q,k)$. However, the last two rows of Table V indicate that such a design is inferior to using the transformed image q. This is because the perturbations introduced by the transformation of image q enhance the robustness of optimized features. We analyze different settings of reference images in Table VI.

TABLE VI

ABLATION STUDY FOR REFERENCE IMAGES. N DENOTES THE NUMBER OF REFERENCE IMAGES

N	mIoU↑	noise↓	recall↑
2	52.29	24.19	74.25
3	53.52	24.02	74.27
4	**53.91**	**23.30**	**75.58**

As shown in Table VI, the performance of CAMs generally improves as the number of reference images (N) increases. The reason is that increasing reference images can generalize land-cover features more comprehensively and provide better

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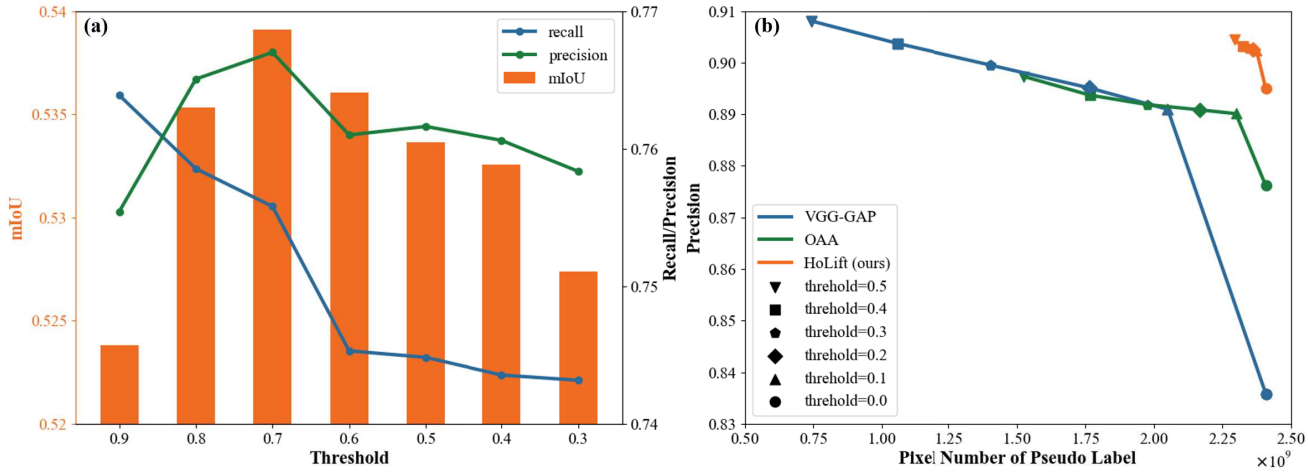


Fig. 6. Ablation study for threshold values on DeepGlobe dataset. (a) Comparison between different threshold values in the training loss L_{cross}. (b) Comparison between different threshold values when generating pseudolabels from CAMs.

TABLE VII
ABLATION STUDY FOR THE PROJECTION HEAD

Projection head	mIoU↑	noise↓	recall↑
	52.02	23.82	75.01
✓	**53.91**	**23.30**	**75.58**

guidance for feature optimization. Therefore, we choose the largest N of 4 within the constraint of our GPU memory. In recent years, researchers in the field of contrastive learning have proposed large and consistent memory banks [47] and queues [46] to enrich reference samples. These designs can store features of abundant samples and slowly update the features during training. However, the activated regions in land-cover CAMs and their image features change rapidly during training [23]. The contradiction compels us to abandon the use of the memory bank and queue.

4) Network Structure: In terms of the network structure, our HoLiftNet employs a simple GAP-based architecture [12] that sequentially performs CAM generation and image classification in a feed-forward pass. Our only modification is the projection head that transforms feature maps for contrastive learning. The numerical results in Table VII show the necessity of the projection head in HoLiftNet.

5) CAM Threshold: Finally, we look into the CAM threshold, which is one of the most important hyperparameters in the whole pipeline of weakly supervised land-cover classification. The CAM threshold is used to locate pixels of each category in the calculation of L_{cross} [see (5)], and it also filters out unreliable pixels from pseudolabels generated from CAMs. Therefore, we perform experiments to determine the best CAM threshold. Different thresholds used in training and inference are compared in Fig. 6(a) and (b), respectively.

As depicted in Fig. 6(a), applying a high threshold of 0.9 produces CAMs with the highest recall. However, using such a high threshold in training severely constrains the extracted regions in CAMs, thereby limiting their feature diversity within each category. Increasing feature consistency between such limited areas by L_{cross} may induce biased feature optimization. As a result, the precision of CAMs and the resulting accuracy of land-cover classification are relatively low. Adopting a lower CAM threshold during training can enrich the features of the extracted regions, facilitating more comprehensive feature optimization. Consequently, the precision improves as the threshold decreases and reaches its peak at the threshold of 0.7. However, the reduction of the threshold also increases misclassified pixels in the extracted regions, especially during the early training epochs. The increase in noise interferes with the enhancement of intraclass feature consistency and leads to a drop in all evaluation metrics. To achieve the highest precision and mIoU score, we employ the CAM threshold of 0.7, which balances the feature diversity and classification precision for the extracted regions.

In the second stage of weakly supervised land-cover classification, generating abundant and accurate pseudolabels from CAMs is fundamental for training a reliable land-cover classification network. Therefore, we compare the pixel number and precision of the pseudolabels generated by different CAM thresholds in Fig. 6(b). We analyze the thresholds from 0 to 0.5, which is an intuitive binary threshold for distinguishing foreground from background. As shown in Fig. 6(b), a lower threshold tends to result in pseudolabels with more pixels. Meanwhile, the precision of pseudolabels declines as more pixels with low CAM responses join the pseudolabels. In terms of both VGG-GAP and OAA, the pseudolabels significantly change with the threshold variation. The contradiction between quantity and quality poses challenges to determining the best threshold. In contrast, our HoLiftNet is insensitive to the CAM threshold for inference. For any threshold value up to 0.5, the pseudolabels generated by HoLiftNet consistently contain abundant pixels with high precision. Our HoLift scheme ensures sufficient pseudolabels by lifting pixelwise responses in CAMs. Furthermore, HoLiftNet improves its capability of identifying land covers by enhancing intraclass similarity and interclass differences, thereby producing pseudolabels with superior precision. According to the results illustrated in Fig. 6(b), the CAM threshold for inference is set to 0.5, which yields pseudolabels with the highest precision and sufficient pixels.

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F. Limitations

In this article, the holistic continuity lifting loss is designed based on the observation that land covers are seamlessly joined without foreground–background distinction. It effectively alleviates discontinuity in pseudolabels by encouraging pixelwise high responses. However, this specialized design is not directly applicable to object-centric semantic segmentation that focuses on distinct foreground targets.

Moreover, our goal is to uncover and exploit hidden cues from image-level land-cover labels to reduce response ambiguity in CAMs. To ensure easy integration with other methods, we focus on developing the training scheme while deliberately avoiding complex network modifications. Consequently, HoLiftNet still faces common challenges like small-target perception and class imbalance. The former arises from detail loss caused by multiple downsampling, which can be addressed by multiscale branches [36] or improved pooling layers [64]. The latter may lead to biased feature learning and CAM responses, which can be alleviated by feature calibration [65] or adaptive thresholding [66].

Although these challenges are beyond the scope of this work, the high extensibility of HoLiftNet allows our training strategy to be integrated with those designs to further enhance performance in such scenarios.

V. Conclusion

In this article, we introduce a HoLift scheme to directly free CAMs of massive ambiguous responses for weakly supervised land-cover classification. With HoLift deployed in a deep image-level classifier, we obtain high-quality CAMs for calculating pseudolabels with the land-cover continuity preserved and the borders clearly delineated on two challenging land-cover classification benchmarks. In-depth ablation studies further verify the effectiveness of every component in HoLift scheme. The qualitative results show how the CAMs are improved with ambiguity eliminated inside land covers and around border areas. Extensive experiments demonstrate that our HoLift scheme enables a state-of-the-art performance for segmenting stuff-level land covers with image-level supervision.

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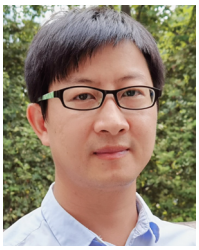
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